# GYANASAI ADUSUMILLI

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## SUMMARY

Data Science master's student at Stevens Institute of Technology (graduating December 2026) who builds end-to-end machine learning systems in computer vision, remote sensing and retrieval-augmented generation (RAG). Industry experience training learning-to-rank models and running A/B tests at OYO; hands-on with PyTorch, scikit-learn, OpenCV and LLM pipelines, from data collection to deployed web demos. Seeking Machine Learning Engineer, AI Engineer and Data Scientist roles.

## EDUCATION

**Stevens Institute of Technology**, Hoboken, NJ — Jan 2025 – Dec 2026 (expected)
*Master of Science in Data Science* | *GPA: 3.66/4.0*

**SRM University AP**, Amaravati, India — Sep 2020 – Jun 2024
*Bachelor of Technology in Computer Science and Engineering*, *Specialization in Artificial Intelligence*

**Relevant coursework:** Machine Learning, Artificial Intelligence, Probability and Statistics, Linear Algebra, Data Structures, Database Management, Parallel Computing, Software Engineering

## TECHNICAL SKILLS

**Languages:** Python, SQL, R, C++, Java, C

**Machine Learning and AI:** PyTorch, TensorFlow, Keras, scikit-learn, Deep Learning, Hugging Face, Sentence Transformers, FAISS, RAG, LLMs (Mistral-7B), NLP, Computer Vision, Learning to Rank, Feature Engineering, A/B Testing, OpenCV, CNNs (ResNet, U-Net), Transfer Learning

**Data and Analytics:** pandas, NumPy, Matplotlib, Spark, Tableau, Power BI, MySQL, MongoDB, Oracle

**Cloud and Tools:** Git, GitHub, Streamlit, Google Earth Engine, Google Colab, AWS, Azure, Linux

## PROFESSIONAL EXPERIENCE

**OYO** | *Technology Engineer Intern* — Jan 2024 – Jul 2024
*Gurgaon, India*

- Drove a **30% increase in user engagement** by launching a mobile-first application that gave users on-the-go access to critical system information.
- Trained **learning-to-rank (LTR) models** with the Ranking and Optimization team to boost high-performing hotels in search results, engineering features from bookings, clicks, ratings, prices, cancellations and location coordinates across **500+ hotels**.
- Evaluated ranking changes through **A/B tests** and analyzed the results with MySQL queries and dashboards, identifying the factors behind higher conversion rates and average booking values.

## PROJECTS

**Water Body Detection from Satellite Imagery** | *PyTorch, U-Net, scikit-learn, Earth Engine* — 2023; rebuilt 2026

- Built a data pipeline that pulls cloud-masked Sentinel-2 and Landsat-8 composites for 8 regions on 4 continents from Google Earth Engine, labeled with same-month JRC surface water data.
- Trained a U-Net CNN that reached **0.984 water IoU and 99.0% pixel accuracy** on Sentinel-2, about two-thirds fewer errors than the NDWI index baseline.
- Benchmarked SVM, random forest, gradient boosting, KNN and logistic regression against the CNN on 500,000 test pixels, using a spatial block split to prevent data leakage.

**Colon Cancer Detection from Histopathology Images** | *PyTorch, ResNet, OpenCV, Streamlit* — 2024; rebuilt 2026

- Fine-tuned an ImageNet-pretrained ResNet-18 with OpenCV CLAHE preprocessing to classify 10,000 H&E tissue images, reaching **100% accuracy and 1.0 ROC-AUC** on 1,500 held-out test images.
- Deployed a public **Streamlit web demo** that serves the model from Hugging Face Hub, with one-click reproducible training in Google Colab.
- Identified data leakage from augmented near-duplicates in the LC25000 dataset and documented a group-aware evaluation plan.

**Symptoma: AI Symptom and Diagnosis Companion** | *RAG, FAISS, LLMs, scikit-learn* — Jun 2025

- Co-built a four-stage pipeline combining ML classification, semantic search and LLM explanations to predict diseases from symptoms.
- Achieved **98% accuracy, precision, recall and F1** with a random forest classifier on structured symptom–disease data.
- Encoded free-text symptoms with Sentence Transformers (all-mpnet-base-v2) and indexed **240,000+ cases** in FAISS; Mistral-7B generated explanations grounded in the retrieved cases.

## ACTIVITIES

- Competed in 5+ hackathons and was part of 2 winning teams building AI-driven environmental solutions.
- Paired 15 upperclassman mentors with incoming students, helping every participant connect with campus resources.